

$$\vec{y} = [y_1, y_2, \dots, y_n]^T \quad (\text{i.i.d})$$

→ θ

$$P(y|\theta) = P(y_1, y_2, y_3, \dots, y_n|\theta)$$

$$= \prod_{i=1}^n P(y_i|\theta)$$

Find θ that makes the observed data most probable.

maximize $\underbrace{P(y|\theta)}_{\text{Likelihood}}$ w.r.t θ .

Maximum Likelihood Estimation:-

$$\begin{aligned} \ell(\theta) &= \log P(\vec{y}|\theta) = \log \prod_{i=1}^n P(y_i|\theta) \\ &= \sum_{i=1}^n \log P(y_i|\theta) \end{aligned}$$

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \ell(\theta)$$

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \sum_{i=1}^n \log P(y_i|\theta)$$

$\hat{\theta}_{MLE} = \arg \min_{\theta} NLL(\theta)$	$\begin{aligned} NLL(\theta) &= -\ell(\theta) \\ &= -\sum_{i=1}^n \log P(y_i \theta) \end{aligned}$
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#1.

$\Rightarrow$ A sequence of n coin tosses
 $\Rightarrow$ Each outcome y_i is a binary R.V. $\begin{cases} \text{Head} = 1 \\ \text{Tail} = 0 \end{cases}$
 $\Rightarrow$ Bernoulli Distribution. $\left| p(\text{Head}) = \theta \right.$

$$p(y_i | \theta) = \theta^{y_i} (1-\theta)^{1-y_i}$$

$$p(y | \theta) = \prod_{i=1}^n p(y_i | \theta)$$

$$\log p(y | \theta) = \sum_{i=1}^n \log p(y_i | \theta)$$

$$= \sum_{i=1}^n \log \left[\theta^{y_i} (1-\theta)^{1-y_i} \right]$$

$$\ell(\theta) = \sum_{i=1}^n \left[y_i \log \theta + (1-y_i) \log (1-\theta) \right]$$

$$\frac{\partial \ell(\theta)}{\partial \theta} = 0$$

$$\frac{1}{\theta} \sum_{i=1}^n y_i + \frac{1}{1-\theta} (-1) \sum_{i=1}^n (1-y_i) = 0$$

$$\frac{1}{\theta} \sum_{i=1}^n y_i - \frac{1}{1-\theta} \left[\sum_{i=1}^n 1 - \sum_{i=1}^n y_i \right] = 0$$

$$\frac{1}{\theta} \sum_{i=1}^n y_i - \frac{1}{1-\theta} [n - \sum_{i=1}^n y_i] = 0$$

$$\frac{1}{\theta} \sum_{i=1}^n y_i - \frac{n}{1-\theta} + \frac{1}{1-\theta} \sum_{i=1}^n y_i = 0$$

$$\frac{(1-\theta) \sum y_i - n\theta + \theta \sum y_i}{\theta(1-\theta)} = 0$$

$$\cancel{\sum y_i} - \cancel{\theta \sum y_i} - n\theta + \theta \cancel{\sum y_i} = 0$$

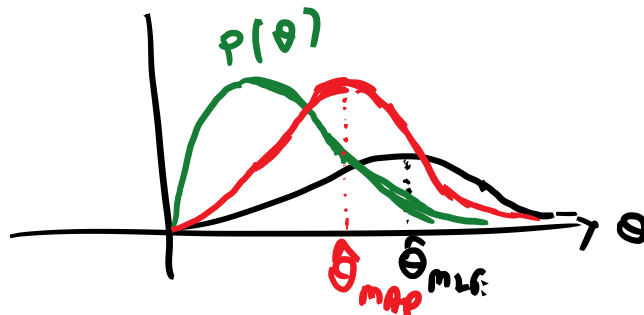
$$\hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n y_i$$

- * If we do not have enough data
- The estimated θ may not be a good fit.
 - It can overfit the data.
 - We need to regularize the problem with some prior knowledge

[we can specify a prior distribution on parameter θ i.e. $P(\theta)$]

- * Uniform prior distribution is the same as using no prior.

$$P(\theta | \vec{y}) = \frac{P(\vec{y} | \theta) P(\theta)}{P(\vec{y})}$$



$$\hat{\theta}_{\text{map}} = \arg \max_{\theta} p(\theta | \vec{y})$$

$$\hat{\theta}_{\text{map}} = \arg \max_{\theta} \log p(\theta | \vec{y})$$

$$= \arg \max_{\theta} \log \frac{p(\vec{y} | \theta) p(\theta)}{p(\vec{y})}$$

$$= \arg \max_{\theta} \left[\log p(\vec{y} | \theta) + \log p(\theta) - \log p(\vec{y}) \right]$$

$$\hat{\theta}_{\text{map}} = \arg \max_{\theta} \sum_{i=1}^n \log p(y_i | \theta) + \log p(\theta)$$

$$\hat{\theta}_{\text{map}} = \arg \min_{\theta} \text{NLL}(\theta) - \log p(\theta)$$

If you choose $p(\theta)$ to be uniform, then $\hat{\theta}_{\text{map}}$ reduces to $\hat{\theta}_{\text{MLE}}$.

Testing a Coin n Times

$$p(y_i | \theta) = \theta^{y_i} (1 - \theta)^{1 - y_i}$$

$\theta \in (0, 1)$, we assume Beta prior.

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$p(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

$\Gamma(\cdot) = \text{Gamma fn.}$

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

$$\Re(z) > 0$$

$$\Gamma(n) = (n-1)!$$

log probability : : : $\hat{\theta}_{MAP} = ?$